Hall Ticket Number:

MR-21

Code: B414CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, Jan/Feb-2024

Computer Organization and Architecture

(Computer Science and Engineering)

Time: 3 Hours

Answer ALL Questions

Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List out the difference between computer organization and architecture.	2	1	2	1
1. b)	Specify various types of instructions based on address?	2	2	1	2
1. c)	Need of addressing mode in an instruction execution?	2	2	1	1
1. d)	Define micro word, micro instruction and micro program?	2	1	1	1
1. e)	Give the flow chart of restoring algorithm.	2	2	5	1
1. f)	Apply 2's complement of the following i) 10010 ii) 111000 iii) 0101010 iv) 111111.	2	3	5	3
1. g)	Classify the cache mapping techniques.	2	2	3	1
1. h)	Explain I/O Bus Vs Memory Bus.	2	2	3	2
1. i)	Explain switch logic behind the implementation of crossbar switch?	2	2	2	5
1. j)	Explain array processors?	2	2	4	5

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Demonstrate the functional units and components that are used for Understanding of computer Organization?	5	2	2	1
2. b)	Design a 4 bit arithmetic combinational arithmetic circuit which implements addition, subtraction, increment, and decrement operations?	5	4	2	2
	OR				
2. c)	Brief discussion about BUS Organization and construction of bus and its operations with neat sketch?	5	2	2	3
2. d)	How to represent an instruction? Explain instruction parts based on bits available in instruction code?	5	1	1	1
3. a)	Write notes on three addresses, two addresses, and one address zero address instructions. And write an assembly language program for x=(a+b) * (c+d) by using two , three and one address instructions?	10	4	1	6
	OR				
3. c)	Explain the general register organization in computer.	5	3	2	1
3. d)	Analyze the functions BUN and BSA instructions.	5	4	1	1
4. a)	What are the steps involved in addition / subtraction of two floating-Point Numbers? Explain the operation through a flow chart. With an example.	6	2	5	9
4. b)	Draw an H/W circuit for fixed point Number ADD/SUB?	4	1	5	11
	OR				
4. c)	Construct a flowchart for 2's Complement multiplication algorithm and Perform multiplication between 10111 * 10011.	10	2	5	4
5. a)	Analyze the hierarchical memory systems are employed to achieve optimal performance by a computer system.	6	4	3	1
5. b)	In a computer, the number of cache hits are 7 and cache miss are 3, then determine the hit ratio of cache memory.	4	5	3	11

P.T.O.

	OR				
5. c)	Sketch the block diagram of DMA in a computer system and discuss the working principle of DMA?	7	2	2	1
5. d)	Explain Hand Shaking method of data transfer?	3	2	2	1
6. a)	Explain the Need of pipeline technique? Implement an arithmetic Pipeline for floating point addition and subtraction.	5	2	4	11
6. b)	Compare and contrast between RISC and CISC computer architectures.	5	2	4	7
	OR				
6. c)	Define cache coherence and why is it important in shared memory Multiprocessor systems? Give any solution to overcome the coherence problem?	7	4	3	2
6. d)	Explain vector instructions and how it will be helpful for performance point of view?	3	3	4	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome